

Cloud Computing Foundations for Data Science

BSIT LECTURE HANDOUT

Course: Data Science Program: BSIT

Infrastructure • Data Pipelines • Analytics Platforms • Security • Cost

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1 Introduction to Cloud Computing

Cloud Computing

Cloud computing is the on-demand delivery of computing services such as servers, storage, databases, networking, analytics, and software over the internet. Instead of buying and maintaining all infrastructure locally, organizations rent resources when needed and pay only for what they use.

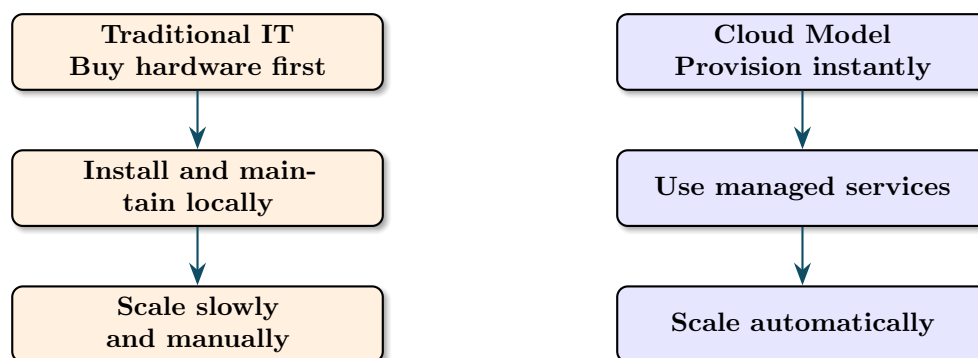
Why It Matters in Data Science

Modern data science depends on large datasets, collaborative tools, scalable compute, and rapid experimentation. Cloud platforms make it possible to train models, store data lakes, deploy dashboards, and serve predictions without building a large physical data center.

1.1 Core Characteristics of Cloud Computing

- **On-demand self-service:** Users provision compute, storage, and analytics resources when needed.
- **Broad network access:** Services are reachable through the internet from laptops, mobile devices, and APIs.
- **Resource pooling:** Shared infrastructure serves multiple customers efficiently through virtualization.
- **Rapid elasticity:** Capacity scales up or down quickly depending on workload intensity.
- **Measured service:** Usage is monitored for billing, governance, and performance management.

1.2 Traditional IT vs. Cloud Model

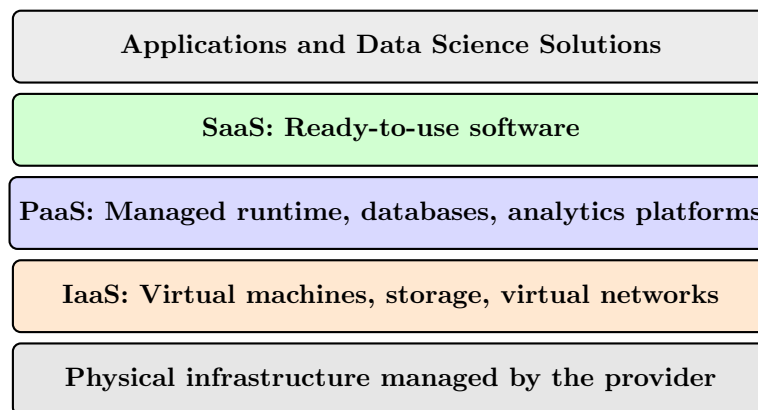


1.3 Why Organizations Move to the Cloud

- Lower upfront capital expenditure.
- Faster deployment of software products and data platforms.
- Better collaboration for distributed teams.
- Easier disaster recovery and backup planning.
- Access to advanced AI, analytics, and managed database services.

2 Service Models and Deployment Models

2.1 Cloud Service Models



Model	What the provider manages	User focus	Examples
IaaS	Virtual machines, networking, disks, load balancers	Operating systems, apps, data, security configuration	Azure VMs, Amazon EC2, Google Compute Engine
PaaS	Runtime, scaling, patching, middleware, deployment platform	Application logic, model code, data products	Azure App Service, Google App Engine, Heroku
SaaS	Complete application	Usage, configuration, access control, data input	Microsoft 365, Gmail, Power BI Service

2.2 Deployment Models

- **Public cloud:** Resources run in infrastructure owned by a cloud provider and shared across tenants.
- **Private cloud:** Cloud-like environment dedicated to a single organization.
- **Hybrid cloud:** Combines on-premises systems with public cloud services.
- **Multi-cloud:** Uses services from more than one public cloud provider.

🔗 Choosing a Model

A university analytics team may keep student records in a private environment for compliance, but train machine learning models in a public cloud GPU environment. This is a typical **hybrid cloud** pattern.

3 Cloud Building Blocks for Data Science

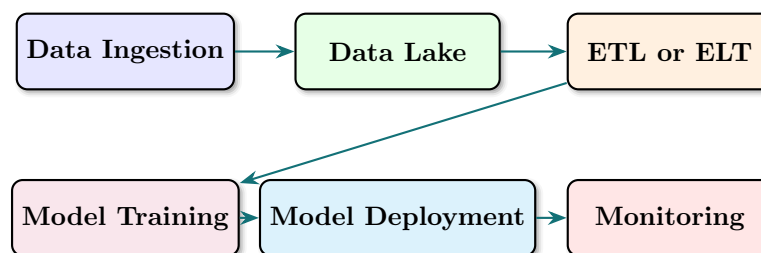
3.1 Essential Infrastructure Components

- **Compute:** Virtual machines, containers, serverless functions, GPU instances.
- **Storage:** Object storage, block storage, file shares, archives.
- **Databases:** Relational, NoSQL, graph, time-series, and data warehouses.
- **Networking:** Virtual networks, subnets, DNS, gateways, firewalls, CDNs.
- **Identity:** Users, service accounts, roles, policies, secrets.
- **Monitoring:** Logs, metrics, alerts, dashboards, tracing.

3.2 Data Science Platform Services

💡 Typical Cloud Services Used by Data Teams

Data science in the cloud often combines storage for raw data, notebooks for exploration, scalable compute for training, data orchestration for pipelines, feature stores, experiment tracking, and managed endpoints for prediction serving.



3.3 Storage Options and Use Cases

Storage Type	Purpose	Data Science Use Case
Object Storage	Cheap, scalable file storage	Store images, CSV files, parquet files, models, notebooks, backups
Block Storage	Attached disk for compute instances	High-performance training workloads on virtual machines
File Storage	Shared directory access	Team collaboration for scripts and intermediate outputs
Archive Storage	Long-term, low-cost retention	Retain old datasets or compliance backups

4 Virtualization, Containers, and Serverless Computing

4.1 Virtualization

📄 Virtualization

Virtualization is the abstraction of physical hardware into multiple virtual machines. A hypervisor allows one physical server to host many isolated computing environments.

- Enables efficient hardware utilization.
- Allows different operating systems to run on the same physical machine.
- Supports isolation, testing, and fast provisioning.

4.2 Containers

Container

A container packages application code together with its dependencies so it can run consistently across environments. Containers are lighter than virtual machines because they share the host operating system kernel.

- Useful for reproducible data science environments.
- Simplifies deployment of APIs, notebooks, and model-serving services.
- Common tools include Docker and Kubernetes.

4.3 Serverless Computing

Serverless

In serverless platforms, developers deploy functions or applications without managing servers directly. The cloud provider scales execution automatically based on demand.

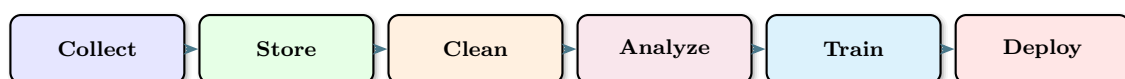
- Best for event-driven jobs, scheduled data processing, and lightweight APIs.
- Pay-per-execution can reduce cost for intermittent workloads.
- Not ideal for long-running, memory-heavy training jobs.

Data Science Example

A serverless function can automatically validate a newly uploaded CSV file, store metadata, and trigger a downstream preprocessing pipeline.

5 Cloud Data Engineering and Analytics Workflow

5.1 End-to-End Workflow



5.2 Common Layers in a Cloud Data Pipeline

- **Ingestion layer:** Collects batch or streaming data from websites, mobile apps, sensors, or business systems.
- **Storage layer:** Data lake or warehouse stores raw and curated data.
- **Processing layer:** Cleans, transforms, enriches, and aggregates data.
- **Analytics layer:** Supports SQL analysis, dashboards, notebooks, and reporting.
- **Machine learning layer:** Trains models, tracks experiments, and serves predictions.
- **Governance layer:** Handles identity, auditing, lineage, security, and cost tracking.

5.3 Batch vs. Streaming Processing

Approach	Description	Example
Batch Processing	Data is processed at scheduled intervals	Nightly sales aggregation, weekly report generation
Streaming Processing	Data is processed continuously as it arrives	Fraud detection, IoT monitoring, clickstream analytics

⚠ Important Design Decision

Not every data science workload requires real-time processing. Streaming adds complexity, so teams should choose it only when low-latency insights create real business value.

6 Security, Governance, and Reliability

6.1 Shared Responsibility Model

📋 Shared Responsibility

In cloud computing, security is a shared responsibility. The provider secures the cloud infrastructure, while the customer secures identities, applications, configurations, data, and access policies.

- Use strong identity and access management.
- Encrypt data at rest and in transit.
- Rotate keys and secrets securely.
- Enable logs, alerts, and regular compliance reviews.
- Design backup and disaster recovery procedures.

6.2 Security Risks in Data Science Systems

- Misconfigured storage buckets exposing sensitive data.
- Weak API authentication for deployed models.
- Insecure notebooks containing secrets.
- Data poisoning or model drift in production pipelines.
- Over-privileged service accounts.

6.3 Reliability Concepts

💡 High Availability and Fault Tolerance

Cloud systems improve reliability by replicating services across zones or regions, balancing traffic, and recovering automatically from hardware failure. For analytics systems, reliability means that pipelines, dashboards, and prediction APIs stay available even when components fail.

6.4 Cost Management Principles

- Shut down idle virtual machines and notebook sessions.
- Use autoscaling instead of overprovisioning.
- Choose storage tiers based on access frequency.
- Monitor data egress and inter-region transfer costs.
- Track experiments to avoid wasteful repeated training runs.

7 Cloud Computing Use Cases in Data Science

7.1 Representative Applications

- **Business intelligence:** Dashboards, KPI reporting, and descriptive analytics.
- **Predictive analytics:** Demand forecasting, churn prediction, and risk scoring.
- **Computer vision:** Image classification, object detection, quality inspection.
- **Natural language processing:** Sentiment analysis, chatbots, document tagging.
- **Recommendation systems:** Product suggestions and personalized content.
- **IoT analytics:** Sensor processing, anomaly detection, and predictive maintenance.

7.2 Mini Case Study: Student Performance Analytics

</> Scenario

A BSIT department wants to predict which students may need academic intervention. Data from enrollment systems, attendance logs, and learning platforms is collected into a cloud data lake. A processing pipeline cleans and joins records, a machine learning model predicts risk scores, and a dashboard shows advisers the students who need support.

Why cloud helps:

- Centralizes data from multiple systems.
- Supports collaborative notebooks for analysis.
- Scales model training during peak academic periods.
- Enables secure dashboards for administrators and advisers.

7.3 Benefits and Challenges

Benefits	Challenges
Elastic scaling for experiments	Cost overruns from poor governance
Fast access to managed services	Misconfiguration and security mistakes
Collaboration across teams and campuses	Vendor lock-in concerns

Disaster recovery and backup options	Need for cloud skills and architectural discipline
Access to GPUs and AI services	Data privacy and regulatory requirements

8 Cloud Skills for BSIT Data Science Students

8.1 Technical Skills to Build

- Linux and command line basics
- SQL and cloud data warehouses
- Python for data pipelines and automation
- APIs and web services
- Containerization with Docker
- CI or CD basics for deployment
- IAM and security fundamentals
- Monitoring and logging concepts

8.2 Professional Mindset

💡 Thinking Beyond Coding

Cloud computing in data science is not only about writing code. It also requires reasoning about scalability, security, maintainability, observability, privacy, and cost. Strong practitioners design systems that are useful, safe, and sustainable.

8.3 Sample Study Roadmap

1. Learn core networking, operating systems, and databases.
2. Practice storing and querying data in a cloud platform.
3. Build a simple ETL pipeline and dashboard.
4. Containerize a machine learning API.
5. Deploy a small end-to-end analytics project with monitoring.

9 Key Takeaways and Review Questions

9.1 Key Takeaways

- Cloud computing provides scalable, on-demand infrastructure and managed services.
- Data science teams rely on cloud platforms for storage, processing, experimentation, and deployment.
- IaaS, PaaS, and SaaS represent different levels of abstraction and management responsibility.
- Security, governance, and cost optimization are part of every cloud architecture decision.
- BSIT students should connect cloud knowledge with data engineering, analytics, and machine learning practice.

9.2 Review Questions

1. What are the five essential characteristics of cloud computing?
2. How do IaaS, PaaS, and SaaS differ from one another?
3. Why is cloud computing useful for data science workflows?
4. When would you choose batch processing over streaming processing?
5. What does the shared responsibility model mean in cloud security?
6. How can poor cloud governance affect the cost of a project?

✎ Short Reflection Task

Design a simple cloud-based data science solution for one problem in your school or local community. Identify the data source, storage layer, analysis tool, model or dashboard output, and one security control you would apply.

End of Handout

Cloud Computing for Data Science — BSIT Program